Ans. (d) :

People	Traits			
	Honest	Ambitious	Intelligent	Industrious
K		✓		✓
L		✓	✓	
M	✓	✓	✓	
N	✓		✓	✓
R	✓			✓

Hence Industrious or ambitious are:
K, L, M, N and R.
Hence there is none in the group who are neither industrious nor ambitious.

127. A man is standing on the 8 m long shadow of a 6 m long pole. If the length of his shadow is 2.4 m, what is the height of the man?

(a) 1.4 m (b) 1.6 m
(c) 1.8 m (d) 2.0 m

Ans. (c) : 8 m long shadow is costed for a height of 6 m.

$\therefore$ 1 m long shadow is costed for a height for $= \frac{6}{8}$m .

$\therefore$ 2.4 m long shadow is costed for a height for $= \frac{6}{8} \times 24$
$= 1.8$m

128. If the angle of a triangle are in the ratio of 4 : 3 : 2, than the triangle:

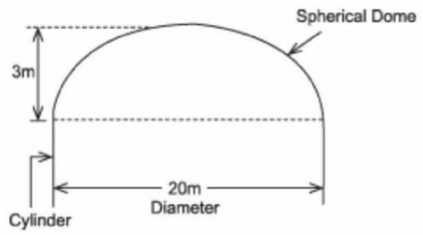
(a) is obtuse
(b) has one angle greater than 80°
(c) is a right triangle
(d) is acute

Ans. (d) : Angles of the triangles may be $4x°$, $3x°$ and $2x°$

$\therefore\ 4x + 3x + 2x = 180 \Rightarrow 9x = 180$
$x = 20°$

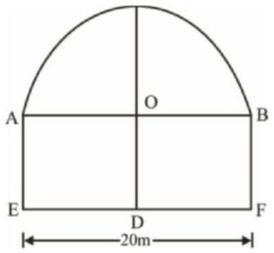
So the angles 80°, 60° and 40°. Hence the triangle will be an acute triangle.

129. The surface area of spherical dome-shaped roof of a cylindrical water tank shown in the figure is:



(a) 60π m^2 (b) 109πm^2
(c) 120π m^2 (d) 300πm^2

Ans. (b) : Let us assume radius of sphere = r m
AO = OB = 10m
OC = 3 m
OD = 2r – 3
$\therefore$ AO × OB = CO × DO
$\Rightarrow$ 10 × 10 = 3 (2r – 3)
$\Rightarrow$ 6r = 109
$\Rightarrow r = \frac{109}{6}$m



$\therefore$ surface area of spherical dome - shaped roof

$$= \frac{4\pi\left(\frac{109}{6}\right)^2 \times 3}{2\left(\frac{109}{6}\right)} = 109\pi m^2$$

130. Strips of two metals A and B are firmly jointed together as shown in the figure.



On heating, A expands more than B does. If this joined strips is heated, then it will appear as:

(a)  (b) 
(c)  (d) 

Ans. (b) : Thermal expansion is the tendency of matter to change in volume in response to a change in temperature. In this case A expands more than B. Therefore, expansion of volume will be more for a in comparison to B on application of same amount of heat. Therefore, on heating it will appear like figure (b).

131. The first feature film (talkie) to be produced in India was:

(a) Hatimatai (b) Alam Ara
(c) Pundalik (d) Raja Harishchandra

Ans. (b) : Alam Ara (1931) was the first Indian feature film (talkie) produced and directed by Ardeshir Irani. The first Indian talkie was so popular that police aid had to be summoned to control the crowds.

132. If X + 2Y = 2X + Y, then X^2/Y^2 is equal to:

(a) 0 (b) 1
(c) 2 (d) 4

Ans. (b) : Given that
$2x + y = x + 2y$
$\therefore\ 2x - x = 2y - y$
$\Rightarrow\ x = y$
$\therefore\ \frac{x^2}{y^2} = \frac{x^2}{x^2} = 1$

133. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Libraries)	List-II (Location)
A. Saraswati Mahal Library	1. Patna
B. Library of Tibetan Work and Archives	2. Dharamshala
C. Raza Library	3. Thanjavur
D. Khuda Baksh Oriental Public Library	4. Rampur

Codes:

	A	B	C	D
(a)	3	2	4	1
(b)	3	2	1	4
(c)	2	3	1	4
(d)	2	3	4	1

Ans. (a) : Library of Tibetan work and archives.

A.	Saraswati Mahal Library	–	Thanjavur Tamil Nadu
B.	Library of Tibetan work and archives	–	Dharamshala Himachal Pradesh
C.	Raza Library	–	Rampur , Uttar Pradesh
D.	Khuda Baksh Oriental public Library	–	Patna Bihar

134. In an office, the distribution of work hours is as shown in the following table:

No. of Staff Members	No. of hours worked
5	0-19
1	20-24
25	25-29
40	30-34
15	35-39
8	40-45

Consider the following inferences drawn from the table:

1. The average number of hours worked by a staff member is about 30.
2. The percentage of those who worked 35 or more hours is less than 25
3. At least 5 staff members worked more than 44 hours.

Which of these inferences is/are valid?

(a) 1 alone (b) 2 alone
(c) 1 and 2 (d) 1, 2 and 3

Ans. (c) : Average no. of hour

$$= \frac{5(17) + 1(22) + 25(27) + 40(32) + 15(37) + 8(42.5)}{5 + 1 + 25 + 40 + 15 + 8}$$

$$= \frac{2957}{94} = 31.45 \sim 30$$

No. of persons who worked 35 or more hours
$15 + 8 = 23$

% of the such persons $= \frac{23}{94} \times 100 = 24.46$
$\Rightarrow < 25$

Hence 1st and 2nd inferences are valid.

135. In a factory quality assurance test is conducted on various samples for a specific characteristic value of the product. The values and the number of samples are as given in the following table:

Characteristic value, X	Number of Samples
10	3
11	7
12	10
13	15
14	28
15	33
16	24
17	11
18	10
19	6
20	3

Consider the following statements based on the table

1. The probability that $X \le 15$ is 0.64
2. The probability that $13 < X \le 17$ is greater than 0.64
3. The probability that $X = 15$ is less than 0.22

Which of the above statements is/are not true?

(a) 1 alone (b) 1 and 2
(c) 2 and 3 (d) 1, 2, and 3